AI Game Theory

Agents with different properties, games with different outcomes?

Different properties could include

- Simulatability (**cooper_characterising_2024**; **kovarik_game_2024**)
- Imperfect Recall (**chen_imperfect_2024**)
- Greater Computational Capacity
- Copyability
- Ability to Commit
- Modifiable Preferences

Commitment

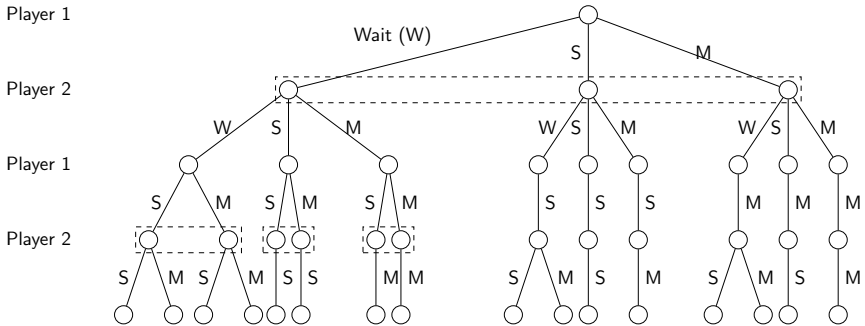
- It's easy to see how commitment could lead to more cooperation.
- But can commitment be used as a threat? And lead to more conflict? (**kokotajlo˙commitment˙2019**)

Game of Chicken

	Straight	**Move**
Straight	$-1, -1$	$1, -\frac{1}{2}$
Move	$-\frac{1}{2}, 1$	$0, 0$

- Two pure Nash equilibria (off-diagonal)
- Mixed equilibrium: Straight w.p. 2/3

Game of Chicken with Commitment



test

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New Equilibrium Outcome

- Outcomes from pure equilibria remain unchanged
- Mixed equilibrium changes

Without Commitment

- Straight with probability $\frac{2}{3}$
- Expected payoff: $-\frac{1}{3}$

With Commitment

- Commit to Straight with probability $\frac{8}{11}$
- Expected payoff: $-\frac{5}{11}$

Commitment can be bad!

Multiple Commitment Periods

	$T = 1$	$T = 2$	$T \to \infty$
P(Straight in round 1)	$\frac{8}{11}$	$\frac{32}{43}$	$\frac{3}{4}$
P(Wait)	$\frac{3}{11}$	$\frac{3}{43}$	0
Expected payoff	$-\frac{5}{11}$	$-\frac{21}{43}$	$-\frac{1}{2}$

This is the worst possible equilibrium!

Extortion Game

- Two players each start with an amount of money M.
- Period 1: Simultaneously choose
  - Keep the money
  - Transfer the money to the other player
- Period 2: Players may punish an opponent who refuses to transfer, but doing so incurs a personal cost c.
- Without commitment, punishment threats are non-credible.
- With commitment, players can commit to punish opponents who do not transfer.
- MSNE: As $T \to \infty$, each player commits to "Keep, Punish" in period 1 with probability $\frac{2M}{2M+c}$
- The expected payoff for each player converges to 0.